

DECLAN KERR

Engineering Portfolio

Mechanical Engineering, University of Connecticut · Class of 2029

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About me

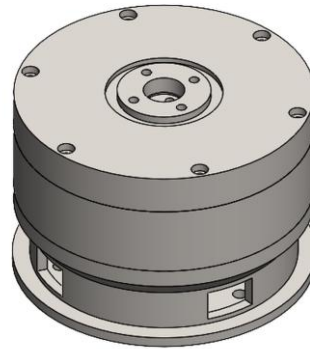
I am a third-year mechanical engineering student at UConn, with experience at an aerospace hydraulic and pneumatic test stand company and in an advanced laser materials processing lab as an undergraduate researcher. I am drawn to projects that bridge mechanical design, electronics, and software, starting with a sketch and ending in a functional device.

Outside of engineering I box competitively, travel as often as I can, and spend time reading, running, and outside with my dog.



Personal Projects

- 8:1 Quasi-Direct-Drive Cycloidal Robotic Actuator (2026)
- Articulating Soldering Fume Extractor (2026)
- Wireless Flow Timer (2025–26)



The assembled actuator on its test rig, and the CAD model it came from.

8:1 Quasi-Direct-Drive Cycloidal Actuator (2026)

About

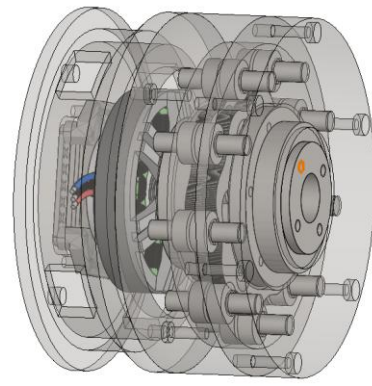
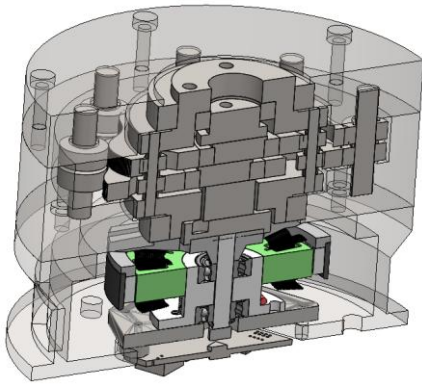
This is an 8:1 quasi-direct-drive actuator I designed and built from scratch, integrating a custom printed cycloidal gearbox with an off-the-shelf BLDC motor and FOC controller. I wanted a robotic joint that would serve as both the foundation for future robotic limb projects and as my introduction to robotics.

QDD actuators are compact, efficient, low-backlash, and most importantly backdrivable: the joint can be pushed back, which lets a limb absorb impact and sense contact through motor current instead of a load cell. The MIT Mini Cheetah popularized the approach and was one of my references for this build. I landed on a desired 8:1 gear ratio as a good balance of speed reduction and backdrivability.

How it works

The motor does not drive the output directly. It drives two eccentric journals, offset 2.0 mm from the axis, each carrying a lobed cycloidal disc. The two discs are clocked 180° apart to cancel each other's imbalanced forces. As the eccentric spins, the disc wobbles inside a ring of nine fixed steel pins. The disc has eight lobes to the ring's nine pins, so every full orbit the mismatch walks the disc backward by exactly one pin: 1/8 of a turn, counter to the input. That is how the 8:1 speed reduction is achieved, all in a compact 40 mm single stage. The reduced and reversed rotation is transmitted to the output pins through the holes in the discs.

The gearbox is a flat puck bolted to the face of the motor that I broke into three subassemblies: the input shaft, ring and output. An adapter mounts the eccentric input stack directly to the rotor face. The ring cup holds the nine Ø6 dowel pins, each carrying two bearings so the disc rolls against the pins instead of sliding on them. Two output plates straddle the disc stack, tying the six output pins together at both ends. The ring lid closes the gearbox and carries the front bearing, and a printed skirt contains the motor and connects the gearbox to the test rig. The motor controller has an integrated encoder that reads the position of a magnet in the rotor and drives it with field-oriented control. The layout is also thermally sound: the gearbox sits axially above the motor instead of wrapping around it and the skirt has windows for cooling.



Half section and transparent view: disc stack between the two output plates, nine ring pins around it, motor below.

Dimensional accuracy was essential since parts press-fit together and any unintended interference shows up as backlash. That meant learning new FDM techniques, like printing test coupons to dial in hole sizes, and buying a filament dryer for the CF-PETG, which is hygroscopic. I also brought code into my CAD workflow for the first time: I used CadQuery to generate several parts, including the cycloidal discs, and built automated geometric auditing into the process, which caught several interferences and errors before they were printed.



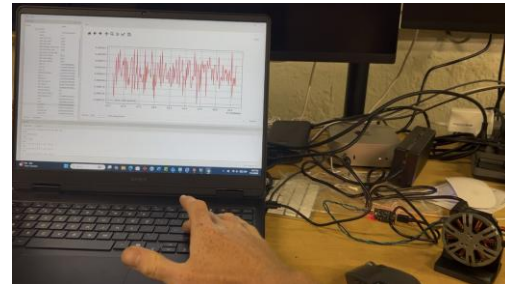
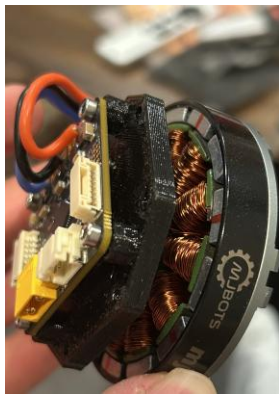
The evolution of the cycloidal discs (left) and ring cups (right). Constant iteration was key.

The assembly runs in four stages. The input stack builds directly on the motor: adapter, riser, eccentric section and disc, spacers, second section clocked 180° with its own bearing and disc, and a far support cap, all clamped by two M3 through-bolts. Then the rear 6807 drops into the ring cup floor, the cup goes over the motor, and the skirt comes up from below. Then the internals: rear output plate into the rear bearing, nine ring pins with their rollers, meshing in the whole input stack, and six output-pin roller stacks onto the rear plate's dowels, all with a healthy amount of grease to limit backlash. Finally, the front plate, front bearing, and lid.



Discs meshed against the ring pins, then the output plate on and the drivetrain closed out.

The moteus-c1 controller board runs field-oriented control over CAN-FD. Bring-up was installing driver and firmware, encoder validation and calibration, then closed-loop position and velocity control on the bench. The motor's torque comes from the controller's current telemetry, using the motor constant and ratio. I found the constant with a bare-motor calibration (shown below) that helped later characterization.



The controller on its factory bracket behind the stator, and closed-loop position control on the bench.

I characterized the actuator using a test rig comprising of a loading arm and a mounting bracket. To measure torque, I hung small weights on the arm at radii of 100, 150, and 200 mm, through holes built into the arm's design, commanded a position hold, then logged steady state current. My peak output torque was ~ 10 N·m, while I was limited thermally to only about 2.5 N·m continuous. I measured backlash, the motion lost at the output with the input held, by rocking the arm and reading the encoder. Efficiency, being mechanical power out over electrical power in, was measured over several load runs. My 54% result was worse than the 60-70% range of my references but I am aware of several parts of the printed gearbox that are contributing to this inefficiency, e.g. a slight misalignment between the inner dowel pins and the front output plate.

Specs and Results

Build and dimensions

- **Cost:** $\sim \$190$ in purchased parts and filament
- **Print time:** ~ 20 h for the actuator, ~ 15 h for the test rig
- **Size:** $\varnothing 112 \times 42$ mm gearbox; $\varnothing 112 \times 73$ mm with motor
- **Mass:** 620 g

Printed parts — carbon-fiber PETG

- Ring cup, ring lid, and housing skirt
- 2× cycloidal disc

- Front and rear output plates
- Eccentric adapter, riser, 2× eccentric section, far support cap
- Ring and output pin spacers
- Test rig and loading arm

Purchased parts

- mj5208 BLDC outrunner motor
- moteus-c1 FOC controller + CAN-FD adapter
- 35 bearings: 2× 6807, 3× 6802, 18× 686, 12× 693
- Steel dowel pins: 9× $\varnothing 6 \times 30$ mm (ring), 6× $\varnothing 3 \times 30$ mm (output)
- M3 and M5 bolts, and 15× heat-set threaded inserts

Performance

- **Reduction:** 8:1, single stage
- **Output torque:** 3 N·m
- **Backlash:** 1.2°
- **Transmission efficiency:** 54%



Articulating Soldering Fume Extractor (2026)

About

Soldering comes with the unfortunate byproduct of toxic flux fumes, so I needed an extraction system if I wanted to safely solder at home. As properly rated commercial options are often \$100–200, I saw the opportunity to build a custom extractor that both saved money and fit my needs. I had a vision of an articulating desk arm that could swap out various useful end effectors for my engineering work: a fume hood attachment, a light, a magnifying glass, a camera, or whatever else I may want. This design allows for maximum mobility in my soldering setup and fume extraction.

A parallelogram-linkage arm clamps to the desk edge and holds a filter box over the workpiece. A fan pulls fumes through a printed hood into a dual-filter stack: an activated carbon pre-filter into a MERV 11-rated filter. The hood attaches magnetically via 4× embedded magnets in both the hood and filter bracket interfaces, snapping easily on and off for filter replacements. Printed knobs with embedded M6 nuts at the linkage joints make the arm tension easily adjustable at any joint, allowing me to set the fume hood facing in any position or orientation that the arm could physically reach. Between the swiveling C-clamp base, the two parallelogram links, and the rotating filter box mount, the arm has four degrees of freedom.



The hood pulled free by hand, and the C-clamp base on the bench edge.

How it works

The C-clamp base is the only part touching the desk and does two jobs: hold against the moment of the arm cantilevered off the edge, and let the whole assembly swivel. A printed screw runs up through the lower jaw to clamp the benchtop, and the vertical post above it carries the first joint.

The arm is made of three parallelogram linkages attached by printed knobs with an embedded nut + screw. The design of all three arms are the same, making it so the arm could theoretically be as long or as short as I wanted. The knobs do well to set friction, loosening when I want to change position and tightening when I want to hold position.

At the working end, the filter box holds the fan, a fan guard, and the two stacked filters, with the printed hood on the front. The box mounts to the arm through its own bracket, and a second bracket ties that to the last arm link – also the interface any future end effector would bolt to.



Arm linkage → generic mount → fan mounting bracket → fan → filter box with fan guard, filter, and pre-filter → hood

The hood is held on by 5 mm neodymium magnets in matched pockets on both parts, which are the perfect strength to hold the fume hood but for an easy detachment by hand. Getting the pocket size right took several iterations: too loose and the magnets would be pulled out of its pocket with its mate, too tight and a press fit would not be possible.



Tension knobs with M6 nuts dropped in mid-print, and learning the importance of supports when printing.

Future upgrades I would consider would be a cleaner routing of the fan cable along the arm, as well as further upgrades to my soldering setup like a custom iron or power supply.



The parallelogram arm off the printer, with the fan, filters, and hardware before assembly.

Specs and Results

Build and dimensions

- **Cost:** ~\$40 in purchased parts and filament
- **Print time:** ~17 h across 15 printed parts

Printed parts (PETG)

- C-clamp base and clamp screw
- Parallelogram arm links
- Tension knobs with embedded M6 nuts
- Filter box and fan spacers
- Filter box bracket and arm mounting bracket
- Hood with embedded magnets and M6 accessory insert boss

Purchased parts

- PC fan
- Activated carbon pre-filter and MERV 11 filter
- 8× 5 mm N-grade neodymium magnets
- M6 bolts and nuts for the linkage joints
- M6 heat-set threaded insert

Performance

- **Degrees of freedom:** 5: swiveling base, three parallelogram links, rotating filter box mount
- **Filtration:** Activated carbon pre-filter for flux VOCs into a MERV 11 particulate filter, rated to capture 65–80% of 1–3 μm particles

Wireless Flow Timer (2025–26)

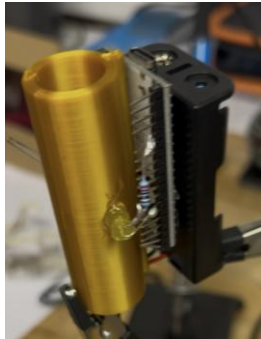
About

This was my first engineering project; I wanted a way to put my new 3D printer and Arduino kit to use.

The problem: measure how long dark liquid takes to clear a length of tube above a shutoff valve. With an expected 3–5 second flow time, a human with a stopwatch would not be accurate enough. My current design uses an optical sensor circuit on the tube to detect the liquid passing, controlled by an ESP32 for the timer and sensing logic, which sends the result over BLE (Bluetooth Low Energy) to an iOS app that logs and displays the data.

How it works

I went through a few detection schemes before landing on optical, primarily a custom valve with a potentiometer on the stem, reading position as the valve opened and closed. What that design doesn't have is clean stopping logic: it depends on whenever the user closes the valve, which is just as inaccurate to flow timing as a stopwatch. An optical sensor, if calibrated properly to the liquid and flow mechanics, is accurate to the hundredths of a second and meshes more cleanly with the iOS app logic.



Intro to soldering thanks to this optical sensing circuit.

Specs and Results

Build

- **Firmware:** C++ on ESP32, written in the Arduino IDE
- **App:** SwiftUI for iOS

Printed parts

- Cylindrical sensor housing locating the optical pair on the tube
- Electronics housing (contains the circuit and epoxy setting)

Purchased parts

- ESP32 dev board
- 5mm LED, phototransistors, wire
- AA batteries and battery holder